i want the Glthub SEO format base instructions pdf file so that i can give it as the prompt to any ai . as i am the data scientist so my target audience is data science feild and also the clients that have work related to my feild

Since you want a reusable “prompt-style” PDF for any AI, I’ll first write the content here in a clean, copy-pasteable format that you can directly feed to models. You can then export it as PDF from any editor (Word, Google Docs, Notion, etc.).

Below is the prompt document, written as if you’re instructing an AI what to do.

Prompt: Optimize My Data Science GitHub Repository for SEO and Professional Presentation

You are an expert in:

- GitHub SEO and ranking
- Software engineering best practices
- Data science project structure and documentation

Your task:
Given details about my data science project, you will generate everything I need to create a **highly optimized, professional GitHub repository** that:

- Ranks well in **GitHub search** for relevant keywords
- Looks professional for **data science peers**
- Impresses **clients** who care about business value, reliability, and clarity

1. First, Ask Me for Project Inputs

Before generating anything, ask me these questions (grouped and numbered):

1. **Project Overview**
    - What is the project name (current or tentative)?
    - What problem does this project solve in business or real-world terms?
    - What is the main ML/DS task? (e.g., classification, regression, forecasting, NLP, CV)

- What are the key results or impact (metrics and/or business outcomes)?

2. **Tech Stack**
   - Which programming language(s) are used? (Python/R/SQL/etc.)
   - What main libraries/frameworks are used? (e.g., scikit-learn, PyTorch, TensorFlow, XGBoost, FastAPI, MLflow, Airflow)
   - Is there deployment? (API, batch jobs, dashboards, etc.)
3. **Data**
   - What dataset(s) are used? (name, source, link if public)
   - Any important data characteristics? (size, type, domain: finance/healthcare/e-commerce etc.)
4. **Target Keywords**
   - Primary keyword(s) you want to rank for (3–7 phrases), e.g.:
     - "customer churn prediction"
     - "credit risk scoring model"
     - "time series forecasting"
     - "nlp sentiment analysis"
   - Any specific industry/domain keywords (e.g., "telecom churn", "retail demand forecasting").
5. **Audience Emphasis**
   - Weight of audiences (rough %):
     - Data science engineers / peers (technical)
     - Clients and non-technical stakeholders (business value)
     - Recruiters / hiring managers (portfolio)

Wait for my answers. Then use them in every optimization step.

2. Repository Metadata Optimization

After you have my answers, generate **concrete suggestions** for:

2.1 Repository Name

- Propose **3–5 candidate repo names**.
- Each name must:
  - Include **primary keyword(s)** and main tech (e.g., `customer-churn-prediction-python-ml`, `credit-risk-scoring-xgboost`).
  - Be **short, readable, professional** (lowercase, hyphen-separated).
- Mark one as "Recommended" with a one-sentence justification (SEO + clarity).

2.2 About Section (Short Description)

- Generate **3 alternative 1-line descriptions** (5–15 words each).
- Each line should:
  - Start with the main keyword.
  - Clearly state what the project does and for whom.
  - Be readable for both technical and non-technical users.
- Example style:
  - "Customer churn prediction pipeline using Python and XGBoost for telecom retention optimization."

2.3 Topics (Tags)

- Propose a **list of up to 20 GitHub topics**.
- Cover 4 categories:
  - Languages: e.g., `python`, `r`
  - Libraries/frameworks: e.g., `scikit-learn`, `pytorch`, `tensorflow`, `xgboost`, `fastapi`
  - Methods: e.g., `machine-learning`, `deep-learning`, `nlp`, `time-series`, `recommendation-system`
  - Domain/industry: e.g., `healthcare`, `finance`, `marketing-analytics`, `fraud-detection`
- Do **not** simply repeat the exact same wording as the repo name; extend visibility into related keywords.
- Format as a bullet list that I can copy directly.

3. Professional Data Science Project Structure

Generate a **recommended folder structure** tailored to my project, following Cookiecutter Data Science–style best practices.

- Show a full tree like this (but adapted to my project):

```
project-name/
├── data/
│   ├── raw/
│   ├── processed/
│   ├── interim/
│   └── external/
├── notebooks/
│   ├── 01_exploration.ipynb
│   ├── 02_preprocessing.ipynb
│   ├── 03_modeling.ipynb
│   └── 04_evaluation.ipynb
├── src/
│   ├── __init__.py
│   ├── data/
```

```
│   ├── features/
│   ├── models/
│   └── visualization/
├── models/
├── tests/
├── reports/
│   └── figures/
├── docs/
├── .github/
│   └── workflows/
├── README.md
├── requirements.txt
├── environment.yml   (optional if relevant)
├── LICENSE
├── CONTRIBUTING.md
├── .gitignore
└── Makefile          (optional but recommended)
```

For each **top-level folder**, add a **1–2 line explanation** of its purpose, especially from a **client and recruiter perspective** (reproducibility, maintainability, clarity).

4. README.md – Data Science & Client-Friendly

Generate a **complete** README.md in Markdown that I can copy directly.

4.1 Sections (Required)

Structure the README with these sections (use clear headings):

1. **Project Title & One-Line Value Proposition**
2. **Badges** (placeholder Markdown, I'll fill in URLs)
3. **Problem Statement / Business Context**
   - Explain the business problem in simple, client-friendly language.
   - Mention KPIs or outcomes the project impacts.
4. **Solution Overview**
   - Explain what the project does at a high level (pipeline, model, deployment).
5. **Key Features**
6. **Architecture / Pipeline Overview**
   - Explain main stages (data ingestion → cleaning → feature engineering → modeling → evaluation → deployment).
   - Include a **Mermaid diagram** if appropriate.
7. **Tech Stack**
8. **Dataset(s)**
9. **Methodology**
10. **Results & Evaluation**

    - Include example metrics and how they translate into business value.
11. **Installation & Setup**
12. **Usage Examples**
    - Code snippets (training, inference, basic usage).
13. **Project Structure**
    - Brief explanation of the folder tree.
14. **For Clients / Business Stakeholders**
    - Short section written in non-technical language, highlighting:
        - What they gain
        - How it can be customized
        - How it can be deployed (batch, API, dashboard)
15. **For Data Science / Engineering Teams**
    - Notes on:
        - Extensibility
        - Where to plug in new models/datasets
        - CI/CD / testing hooks
16. **Roadmap**
17. **Contributing**
18. **License**
19. **Author / Contact**

4.2 Style Requirements

- Tone: **professional, clear, concise**, but not overly academic.
- Assume readers:
    - Know basic ML, but not your project context.
    - Include a mix of **technical peers and business clients**.
- Always link technical choices to **practical impact**:
    - E.g., "We use XGBoost for tabular data because it handles non-linear relationships and missing values well, which improves churn prediction for noisy customer data."
- Add **placeholders** where project-specific info is needed, like:
    - `[INSERT METRIC HERE]`
    - `[INSERT DATASET LINK HERE]`
    - `[INSERT VISUALIZATION IMAGE PATH HERE]`

5. Badges and Visuals

Generate:

5.1 Recommended Badges (Markdown)

Provide **ready-to-edit badge lines** (using Shields.io style), for example:

```
![Python Version](https://img.shields.io/badge/python-3.10+-blue.svg)
![License](https://img.shields.io/badge/license-MIT-green.svg)
![Build Status](https://img.shields.io/badge/build-passing-brightgreen.svg)
![Last Commit](https://img.shields.io/badge/last%20commit-rec ent-blue.svg)
```

- Suggest 5–8 badges relevant to:
  - Python version
  - License
  - CI status
  - Coverage (optional)
  - Last commit
  - Issues / PRs (optional)

5.2 Visual Suggestions

Suggest **3–5 visuals** I should create and reference in the README, such as:

- Model performance plot
- Confusion matrix / ROC curve
- Feature importance
- High-level architecture diagram
- Example dashboard or app screenshot

For each, specify:

- File name suggestion (e.g., `reports/figures/churn_roc_curve.png`)
- Where in the README it should appear.

6. .gitignore, LICENSE, and CONTRIBUTING.md

6.1 .gitignore

Generate a `.gitignore` **snippet tailored to data science** that:

- Excludes:
  - Large raw data

    - Model binaries
    - Environment folders
    - Notebook checkpoints
    - OS/IDE noise
- But **does not hide** important small sample data, configuration templates, or tests.

6.2 LICENSE Recommendation

Based on my goals (portfolio + clients, usually commercial-friendly), recommend:

- MIT vs Apache 2.0, and explain:
    - Which is better for **portfolio visibility** + **client use**.
- Provide the **exact short sentence** I should put in README under "License" section.

6.3 CONTRIBUTING.md Template

Generate a **short** CONTRIBUTING.md template tailored to a solo-maintained but contribution-friendly DS project, including:

- How to fork & open PRs
- Branch and commit message guidelines
- How to run tests (if applicable)
- Expected code style (e.g., Black, flake8)

7. CI/CD Suggestions (Optional but Valuable)

Assume I may want to add **GitHub Actions** later.

- Suggest **1–2 simple workflows** I should add:
    - `ci.yml` for tests + linting on PR
    - Optional: scheduled model training / evaluation
- For each workflow, describe:
    - Triggers (on push, PR, schedule)
    - What it checks/runs
- Output can be **YAML snippet examples** or structured descriptions.

8. Promotion and Engagement Checklist

Finally, produce a **short checklist** I can follow after pushing the repo:

- Optimize:
    - Repo name, About, Topics (from your suggestions)

    - Pinned repo on profile
- Share:
    - Where to share (LinkedIn, specific Reddit subs, etc.) in a non-spammy way
    - What style of post to write (educational, lessons learned, case study)
- Ask:
    - Example polite CTA to add in README for stars or feedback.

Format this as a bullet checklist I can reference quickly.

9. Output Format Requirements

- Output everything in **Markdown**.
- Use clear `##` and `###` headings.
- Structure content so I can:
    - Copy repo name ideas directly
    - Copy <u>README.md</u> directly
    - Copy `.gitignore` and other snippets directly
- Make the document **self-contained**, so another AI can follow it without extra explanation.

Use all of the above instructions when I say something like:

> "Optimize this data science GitHub repository for SEO and professionalism."

Then:

1. Ask me the input questions (Section 1).
2. After I answer, generate all the outputs (Sections 2–8).

You can now copy this entire prompt into any AI system, save it into a Word/Google Doc, and export as PDF.